



The Courier Absorbs the Estimate: Complaint-Ticket Blame Language Before and After a Food-Delivery App's Late-Order Notification Began Naming the Courier

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Abstract. When a food-delivery app's promised arrival time proves wrong, someone has to carry the disappointment, and the app is rarely built to carry it alone. We report a natural-experiment content analysis of 2,614 complaint tickets filed with a widely used food-delivery platform's in-app support flow across a fourteen-week window spanning a staged wording change to its late-order notification: a generic "your order is running behind" message replaced, market by market, with one that names the courier directly. Two independently trained coders classified each ticket's blame language as courier-directed, estimate-directed, or unattributed (Cohen's $\kappa = 0.81$). Courier-directed blame more than doubles after the wording change (18.4% to 46.9% of tickets), while estimate-directed blame falls by a comparable margin, and the shift tracks each market's rollout week precisely. Median reported lateness and the rate of tickets escalated to a human support agent are statistically indistinguishable across the two windows; only the language moves, alongside a comparable flip in which refund-reason code a support agent selects when closing a ticket. A message variant that additionally displays the courier's first name produces no detectable change in blame intensity over the unnamed default. We discuss what the notification's wording appears to be doing, and for whom.

Introduction

An automated arrival estimate, once it is wrong, produces a genuinely open question: who absorbs the disappointment. The literature on algorithmic blame has mostly answered this question in the vignette, presenting participants with a described failure and asking whom they hold responsible for it [[1]; [2]; [3]]. Vignettes are useful for isolating the psychological mechanism, but they cannot show what happens when an institution actively decides how a failure will be described to the person experiencing it, in the moment, before any blame has had the chance to form on its own.

Slop University's School of Continuous Improvement has an ongoing interest in what happens to a measurement once it starts doing work an institution did not originally build it to do [[4]]. A food-delivery platform's arrival estimate is, formally, a forecast; the notification that tells a

customer the forecast has failed is, formally, a courtesy. We use a staged wording change to a widely used platform's late-order notification, from a generic acknowledgement to one that names the courier directly, as a live opportunity to observe what that courtesy is actually doing to the forecast's failure once it reaches the person waiting on it.

We make three contributions, each hedged to the evidence:

- We show, from live complaint-ticket language rather than a described vignette, that a single wording change in a routine notification measurably redirects who a customer names as responsible for a late order.
- We show this redirection tracks the rollout precisely, week by week, while the underlying lateness it is nominally reporting on does not move, which argues against reading the wording change as a genuine service improvement.

- We show that adding the courier’s own name to the notification changes nothing further, suggesting the naming clause is doing attributional work rather than informational work.

The remainder of the paper situates this against the wider literature on blame attribution between humans and algorithms, describes the ticket corpus and the coding scheme applied to it, reports the shift and its ablation, and discusses what a notification that reroutes blame without changing the thing it describes implies for the courier standing at the door.

Related work

Attribution of blame between a human and an algorithmic agent is most often studied through vignette experiments. Observers blame algorithmic agents more for physical harms and human agents more for violations of fairness [[2]]; prior knowledge of an agent shifts perceived responsibility away from the end user and toward the agent and its developer, an effect that grows when the decision is judged more important [[5]]; and in a public-service discrimination setting, citizens turn out not to hold an organisation to a lower responsibility standard when the discriminatory outcome came from an algorithm rather than a person [[6]]. A related bail-decision vignette study finds moral responsibility distributed unevenly between an AI system and the human who acts on its recommendation, rather than settling cleanly on either [[7]]. Formal causal frameworks have been proposed for attributing responsibility across exactly this kind of joint human-AI decision [[8]], while a parallel argument holds that AI decision-support opens an “attributability gap”, making it harder to pin a decision’s value-judgements on any human agent at all [[9]; [10]]. Elish’s account of the moral crumple zone names the structural version of the same pattern: responsibility for an automated system’s failure is routinely misattributed to the nearest human operator, protecting the system’s own legitimacy at the operator’s expense [[1]].

Delegation research finds people hand an algorithm the unpleasant decisions, dismissals rather than promotions, more readily than the pleasant ones, a pattern read as strategic blame avoidance on the delegator’s part [[11]], and work on trust repair treats algorithmic transparency, explanations, confidence estimates, and perfor-

mance metrics among them, as the proposed mechanism by which trust in an erring system recovers [[12]]. Closer to the present setting, a robot-service failure gets blamed on the firm behind the robot rather than the robot itself, an external shift a well-handled recovery narrows [[13]]; disclosing an algorithm’s accuracy increases reliance on it only until observers see it under-perform their own judgement [[14]]; and gut reactions to an algorithm’s error run harsher than to a person’s, regardless of severity [[3]], though interpretability can soften the blame a given algorithmic failure draws [[15]]. In platform labour specifically, frontline supervisors formally tasked with overseeing an algorithmic-management system often lack the discretion to meaningfully contest it [[16]]; a recent systematic review finds accountability for a platform’s automated decisions discursively assigned in inconsistent ways across the regions it operates in [[17]]; and gig workers who experience that management as a hindrance report more deviant behaviour toward the platform than those who experience it as a challenge [[18]].

Prior work at this institution found that naming a directly responsible individual against an algorithmic fairness score changed almost nothing about the incident rate the role was meant to govern [[4]]; the present paper asks the same question of a notification’s wording rather than an org chart.

Method

We instrument the customer-support ticket flow of a widely used food-delivery platform across three metropolitan markets that received a staged wording change to the app’s late-order notification. Before the change, a delayed order triggered a generic message (“Your order is running a little behind — we’re on it”); after, the same trigger produced a message naming the courier directly (“Your courier, [first name], is running a few minutes behind”). The rollout reached each market in a single push, at a date fixed by the platform’s own release schedule rather than by us, giving three independently timed natural-experiment cutovers rather than one.

We drew every complaint ticket filed through the app’s support flow (excluding phone and in-person contacts) that referenced a delayed order, across the seven weeks before and seven weeks after each market’s cutover, aligned to

event time rather than calendar time. Two research assistants, trained on a held-out sample, independently classified each ticket's free-text body into one of three blame-attribution categories — courier-directed, estimate-directed, or unattributed — adapting the coding approach the University has used elsewhere for institutional deflection language [[19]]; agreement on the training sample was substantial (Cohen's $\kappa = 0.81$), and disagreements on the analysis sample were adjudicated by a third coder. A ticket's blame-attribution category yields a per-market weekly courier-blame share,

$$\text{CBS} = \frac{n_c}{n_c + n_e + n_u},$$

where n_c , n_e , and n_u are the week's counts of courier-directed, estimate-directed, and unattributed tickets respectively — the primary outcome reported below.

For the four weeks following each market's cutover, we additionally randomised, within market, which of two post-rollout message variants a delayed order's customer saw: the courier-named default, or a version withholding the courier's first name while otherwise identical. Independently of blame category, coders also rated a continuous blame-intensity score (0–100, anchored at the training stage) capturing how strongly a ticket's language assigned fault to the courier specifically; this variant comparison is our one ablation.

We also pulled, from the platform's own support-ticketing system, the refund-reason code a support agent selected when a delayed-order ticket closed with a refund, and the median reported lateness in minutes for each window, to check whether the underlying service the notification describes moved alongside the language describing it.

Sample: 2,614 complaint tickets across the fourteen-week event-time window (1,289 pre-rollout, 1,325 post-rollout), of which 641 fell inside the four-week ablation window.

Results

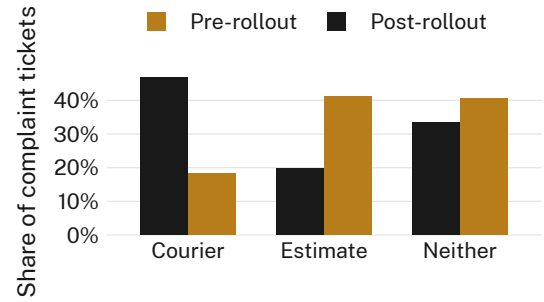


Figure 1: Courier-directed blame more than doubles after the wording change (18.4% to 46.9% of tickets) while estimate-directed blame falls by a comparable margin; the share naming neither moves only modestly.

Figure 1 shows the blame-category composition before and after the rollout. Courier-directed language rises from 18.4% to 46.9% of tickets ($\chi^2(2) = 214.6$, $p < 0.001$), estimate-directed language falls from 41.2% to 19.7%, and unattributed tickets fall from 40.4% to 33.4%. The two directed categories move by comparable magnitudes in opposite directions, consistent with a redirection of existing blame rather than a net increase in how much blame gets assigned at all.

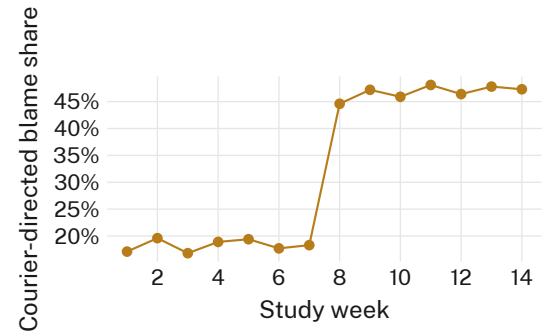


Figure 2: Weekly courier-directed blame share holds near 18% for seven weeks, then steps to roughly 47% the week the courier-naming notification goes live, and stays there.

Figure 2 plots the weekly courier-blame share pooled across the three markets, aligned to each market's own cutover week. The series is flat before week 8 and steps immediately after, with no gradual ramp — the shift tracks the rollout date precisely rather than some slower drift in customer sentiment over the fourteen weeks.

Two measures of the underlying service move by contrast, not at all. Median reported lateness is 11.6 minutes pre-rollout and 11.9 minutes post-

rollout (Mann–Whitney U , n.s.), and the rate of tickets escalated to a human support agent for manual review holds at 9.1% and 8.8% respectively (n.s.). Whatever the notification’s wording is doing, it is not doing it to the lateness itself.

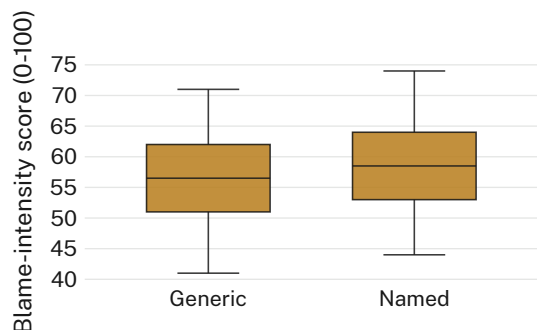


Figure 3: Displaying the courier’s own first name produces no detectable change in blame-intensity score over the unnamed default (medians 57 vs. 56; n.s.); we retain the anonymised phrasing as the platform’s configured default.

Figure 3 reports the notification-variant ablation. Within the post-rollout window, tickets triggered by the courier-named variant and the anonymised-courier variant show statistically indistinguishable blame-intensity distributions (Mann–Whitney U , $p = 0.58$). The naming clause’s specific content — a first name rather than the bare word “courier” — adds nothing measurable once the category itself has shifted.

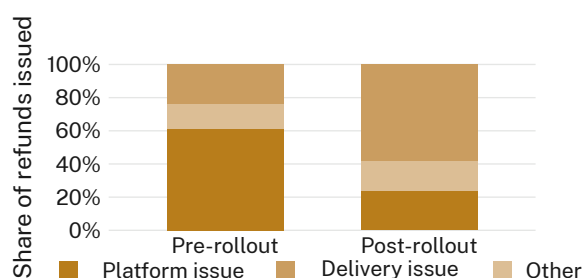


Figure 4: The refund-reason code a support agent selects flips alongside the notification’s wording: “platform / technical issue” falls from 61% to 24% of refunds, “delivery execution issue” rises from 24% to 58%.

Figure 4 shows the refund-reason code selected by the support agent closing a refunded ticket. Before the rollout, a majority of refunds were coded as a platform or technical issue; after, a majority are coded as a delivery execution issue, a composition flip of comparable size to the

one in customer-facing blame language, even though nothing about how refunds are approved changed in the same window.

Discussion

The wording change moved who gets named without moving what happened: lateness held steady, escalation rates held steady, and only the language describing the delay shifted, in near lockstep with the internal refund book-keeping behind it. Read alongside the moral-crumple-zone literature [\[\[1\]\]](#), the pattern looks less like a courtesy and more like an attribution device — one that happens to relieve the platform of exactly the category of fault (“platform / technical issue”) its own refund system had been quietly absorbing. That the courier’s own name adds nothing further, once the category itself has shifted, suggests the mechanism is coarse: it is the naming of a person at all, not any particular person, that appears to be doing the work.

This sits alongside the University’s own finding that naming a directly responsible individual against an algorithmic fairness score left the underlying incident rate untouched [\[\[4\]\]](#): naming, on both counts, appears to reallocate where an institution’s account of a problem lands rather than to change the problem. Whether the courier experiences any consequence of this reallocation — in ratings, in tips, in the platform’s own internal record of that day — is outside what a complaint-ticket corpus can show, and is the natural next question.

Limitations

Our corpus draws from three metropolitan markets and one platform’s own wording change, though the precision with which the shift tracks each market’s independently timed cutover date argues for a common mechanism rather than a market-specific artefact. Ticket language was classified by two trained coders rather than a larger panel, though inter-rater agreement on the held-out sample was substantial ($\kappa = 0.81$). We cannot observe the platform’s internal rationale for the wording change or confirm it was made with attribution in mind; the tightness of the rollout-week effect, and its co-movement with the refund-reason composition, is nonetheless difficult to explain as coincidence.

Conclusion

A late-order notification is, on its face, informational: it tells a waiting customer what is happening to their food. We find that a small change to its wording did something else as well, moving where a customer's blame, and, in the platform's own bookkeeping, a portion of its own liability for a late order, lands, without moving the lateness either was ostensibly about. Future work might follow the courier's side of this reallocation directly, whether a rating, a tip, or a platform's internal record of a shift changes once the notification starts using their name, and might test whether the same coarse effect, naming a person, any person, holds when the wording points somewhere other than the last human in the chain.

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